Chapter 19 exercises • Windows 7

OS19-E01 • Granted rights and SAME_ACCESS

A handle0x40 grants abstractR1, notW2. Successful DUPLICATE_SAME_ACCESS gives B handle0x88.

1. Can either write?
2. Must B's0x40 name the same object? Does A close freeF?
3. Does this subset model describe every different-access DuplicateHandle?

OS19-E02 • Session scope

S1/S2 have interactive users and Local\Ready events. Authorized Global\Ready access is also possible.

1. Which resources are separate/shared?
2. Does equal Local name mean equal object? Does Global bypass DACL?
3. Are these two VMs?

OS19-E03 • Actual ReadFile outcomes

Windows 7 regular fileABCDEFGHIJ, FILE_FLAG_OVERLAPPED, requests 6 at offsets0/6/10.

1. Interpret FALSE/ERROR_IO_PENDING. May lpNumberOfBytesRead be NULL?
2. Give final bytes/data and required lifetimes.
3. Does successful CancelIoEx permit immediate free?

OS19-E04 • IRQL and thread priority

A DISPATCH_LEVEL DPC wants a 5 ms I/O wait and potentially faulting paged-pool access while another CPU updates a shared list.

1. Are these operations legal? Define alternative work placement.
2. Does raising local IRQL exclude that CPU?
3. Why does priority31 not guarantee a hard deadline?

OS19-E05 • Fixed priority queues

P1/P2 priority8, ready0, bursts4/3. P3 priority10, ready1, burst2. Equal q2, no boosts/decay, higher priority preempts immediately;preempted task retains residual slice/front position.

Chapter 19 exercises • Windows 7 (continued)

1. Trace intervals/completions.
2. Which premise fails if P1 continues until2 after P3 arrives1?
3. Is q2 universal Windows 7 behavior?

OS19-E06 • Reserve, commit, resident

Private base0x100000, reserve64 KiB, commit two4 KiB pages 0x101000/0x102000, not yet touched.

1. Give three quantities before/after touching0x101123.
2. Decommit first page, then release region. What changes?
3. Does commit immediately zero-write both pagefile pages?

OS19-E07 • Ordered DACL

Alice belongs to Staff. AbstractR1/W2. ACEs allowAliceR, denyStaffW, allowStaffRW. Exclude other policy/privilege rules.

1. Trace remaining forRW.
2. Move allowStaffRW before deny. What happens?
3. Are NULL and empty DACL equivalent?

OS19-E08 • NTFS extents

Cluster4096 B, disk volume base1048576. Extents(VCN0, len4, LCN100),(4,3,200),(7,2, hole),(9,2,190).

1. Map file byte20603 toLCN/volume/disk byte.
2. Read VCN7+123 and calculate allocation for 11 logical clusters.
3. What do MFT resident attributes and sequence numbers mean?

OS19-E09 • Single-transaction WAL

Old:size 4096, map[100], bit200=0. New:8192,[100,200], bit1. Complete old/new records persist first;steal/no-force are allowed.

1. No commit, only home bit became1. Recover it.
2. Durable commit, all home still old. Recover it.
3. Do checkpoints/metadata recovery guarantee all user data?

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Chapter 19 exercises • Windows 7 (continued)

OS19-E10 • Snapshots and remote uncertainty

LiveX41/Y80 createsS;X changes42 then 43. Remote append is sent but client times out.

1. Give live/snapshot values and old-backing reclamation condition.
2. Is S necessarily application-consistent and an independent backup?
3. Is retry guaranteed not to duplicate append?

References and attribution

Abraham Silberschatz, Peter B. Galvin, Greg Gagne. Operating System Concepts, 9th Edition.

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